

# 1. LITERATURE SURVEY

Recent studies have focused on improving object tracking in autonomous systems using sensor fusion techniques. The Extended Kalman Filter has been widely used to handle nonlinear measurements from radar and lidar sensors for accurate state estimation. Several researchers have demonstrated that combining multiple sensors significantly enhances tracking performance compared to single-sensor approaches. Furthermore, the application of Kalman filtering methods enables real-time prediction and correction of object motion in dynamic environments. These approaches are particularly effective in self-driving systems for estimating position and velocity with reduced noise and improved accuracy.

## 2. Introduction and Background

Autonomous driving systems are rapidly advancing due to their potential to improve transportation safety, efficiency, and reliability. A critical component of these systems is accurate object tracking, which enables vehicles to detect and respond to dynamic obstacles in real time. However, real-world environments introduce challenges such as sensor noise, uncertainty, and nonlinear motion patterns.

To address these challenges, sensor fusion techniques are widely used to combine data from multiple sensors such as radar and lidar, providing a more reliable and comprehensive understanding of the environment [2][5]. The Extended Kalman Filter (EKF) plays a significant role in handling nonlinear measurement models and estimating the state of moving objects with improved accuracy [4][8].

This literature survey specifically focuses on the application of Extended Kalman Filter-based sensor fusion techniques for object tracking in autonomous systems, with an emphasis on improving estimation accuracy, handling nonlinear dynamics, and ensuring real-time performance.

## 3. Problem Statement

In autonomous driving systems, accurate object tracking is a critical challenge due to the presence of noise, uncertainty, and nonlinear behavior in sensor measurements. Although sensor fusion techniques using radar and lidar improve detection accuracy, existing approaches still face limitations in handling nonlinear dynamics efficiently in real-time

environments. Traditional Kalman Filter methods are not suitable for nonlinear systems, while Extended Kalman Filter (EKF) approaches introduce approximation errors due to linearization. Therefore, there is a need for a more robust and efficient state estimation method that can improve tracking accuracy while maintaining real-time performance in dynamic conditions.

## 4. Research Objectives

- To analyze existing sensor fusion techniques used in autonomous systems.
- To study the performance of Kalman Filter and Extended Kalman Filter for object tracking [4].
- To identify the limitations of current tracking methods in nonlinear environments.
- To improve state estimation accuracy using Extended Kalman Filter.
- To ensure real-time performance in dynamic and noisy conditions.
- To develop an efficient sensor fusion approach using radar and lidar data [2].

## 5. Methodology of Literature Review

The methodology adopted for this literature survey follows a systematic and structured approach to ensure comprehensive coverage and reliability of the selected studies.

Relevant research papers were collected from credible sources such as Springer, ScienceDirect, IEEE Xplore, and MDPI databases [2][8]. The selection of papers was based on keywords including sensor fusion, Kalman Filter, Extended Kalman Filter, object tracking, and autonomous systems.

Inclusion criteria focused on recent studies related to radar and lidar-based tracking, nonlinear state estimation, and real-time implementation of filtering techniques [4]. Papers that lacked experimental validation or were not directly related to object tracking were excluded from the review.

The selected studies were analyzed and compared based on key performance metrics such as accuracy, robustness, computational efficiency, and real-time applicability [3]. Special emphasis was given to methods that integrate multiple sensors, as they demonstrate improved performance over single-sensor approaches [5].

This systematic methodology ensures transparency, reproducibility, and a clear understanding of existing techniques, helping to identify limitations and potential areas for improvement in sensor fusion-based tracking systems.

A total of 10 peer-reviewed research papers were selected from reputed databases such as Springer, IEEE Xplore, ScienceDirect, and MDPI to ensure a reliable and comprehensive literature analysis [2][8].

## 6. Review of Existing Work

Recent research in autonomous systems has extensively explored sensor fusion techniques to improve object tracking accuracy and robustness. Studies demonstrate that integrating radar and lidar data provides complementary advantages, where radar offers reliable velocity and distance measurements under adverse weather conditions, while lidar delivers high-resolution spatial information for precise object localization [2][5].

The Kalman Filter has been widely adopted for state estimation due to its computational efficiency and recursive nature. However, its limitation to linear system models restricts its applicability in real-world scenarios, where system dynamics and sensor measurements are often nonlinear [4]. To address this limitation, the Extended Kalman Filter (EKF) has been proposed, which linearizes nonlinear models using Jacobian matrices, enabling more accurate state estimation in dynamic environments [8].

Several studies highlight that EKF-based sensor fusion significantly improves tracking performance by continuously predicting and updating object states such as position and velocity [3][9]. Compared to single-sensor approaches, multi-sensor fusion reduces uncertainty and enhances robustness against noise and environmental disturbances. However, the performance of EKF is highly dependent on proper tuning of process and measurement noise parameters, which directly impacts estimation accuracy.

Furthermore, recent works emphasize the trade-off between computational complexity and estimation accuracy. While EKF provides improved performance for nonlinear systems, it introduces additional computational overhead due to linearization and matrix operations. This makes real-time implementation challenging in highly dynamic environments, especially when multiple objects are tracked simultaneously [6][7].

Overall, existing research establishes that EKF-based sensor fusion is an effective approach for object tracking in autonomous systems, but challenges related to parameter tuning, computational efficiency, and handling highly nonlinear conditions still remain.

## 7. Critical Analysis and Comparison

The Kalman Filter is efficient and fast but is limited to linear systems, making it less suitable for real-world applications [4]. The Extended Kalman Filter (EKF) improves performance by handling nonlinear models, but it introduces approximation errors and requires careful tuning [8].

Sensor fusion using radar and lidar enhances tracking accuracy by combining complementary data, where radar works well in harsh conditions and lidar provides high-resolution information [2][5]. However, issues such as noise, environmental effects, and computational complexity still remain.

| Method                        | Advantages                                                                       | Limitations                                                                |
|-------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Kalman Filter                 | Fast computation, simple implementation, efficient for real-time systems         | Limited to linear models, poor performance in nonlinear conditions         |
| Extended Kalman Filter (EKF)  | Handles nonlinear systems, improved estimation accuracy, widely used in tracking | Linearization error, requires careful parameter tuning, higher computation |
| Sensor Fusion (Radar + Lidar) | High accuracy, robust in different environments, reduces uncertainty             | High computational cost, complex integration, expensive sensors            |

Among the methods, the Extended Kalman Filter provides a balance between accuracy and computational efficiency, making it suitable for real-time applications [8]. However, its performance strongly depends on proper parameter tuning and system modeling.

## 8. Research Gaps

Despite significant advancements in sensor fusion and Extended Kalman Filter-based tracking, several limitations still exist. Most existing studies focus on improving accuracy but do not address real-time computational efficiency in complex environments. Additionally, the

issue of parameter tuning in EKF is not fully optimized, which affects performance under varying conditions.

Furthermore, limited attention has been given to handling highly dynamic and noisy environments effectively. These gaps highlight the need for developing a more robust and efficient sensor fusion approach for accurate and real-time object tracking.

## **9. Alignment to Proposed Work / Project**

The reviewed literature provides a strong foundation for the development of the proposed project on object tracking using sensor fusion. The limitations identified in existing methods, such as nonlinear estimation challenges and parameter tuning issues in EKF, directly influence the design of this project [8].

Based on the analysis, the proposed system focuses on improving tracking accuracy by effectively combining radar and lidar data using the Extended Kalman Filter [2][5]. The study of existing approaches helps in selecting appropriate models, tuning parameters, and ensuring real-time performance.

Thus, the insights gained from the literature survey are directly applied to design a robust and efficient tracking system that addresses the identified research gaps.

## **10. Concepts of Kalman Filtering**

### **10.1 Concepts**

- The Kalman Filter estimates the state of a system using noisy sensor data [2][5].
- It uses mathematical models to represent system behavior and predict future states [4][7].
- The filter works recursively, meaning it updates results continuously over time [3][9].
- It combines prediction and measurement to produce more accurate results [6][10].

### **10.2 Rules**

- The prediction step estimates the next state based on the current state and system model [2][4].
- The update step corrects the predicted state using new sensor measurements [8][10].
- The Kalman Gain determines how much weight is given to prediction vs measurement [3][9].
- The process repeats continuously for real-time tracking [5][7].

### **10.3 Principles**

- Error minimization
- Gaussian noise
- Linear algebra
- Recursive update

## 10.4 Laws

- Probability law
- System dynamics
- Least squares
- Recursive estimation.

## 11. Mathematical Equations and Relationships in EKF

The Extended Kalman Filter uses mathematical equations to estimate the state of a nonlinear system by combining prediction and update steps. The prediction equations estimate the next state and covariance using the system model, while the update equations correct the prediction using sensor measurements [\[2\]](#)[\[4\]](#).

### 11.1 Prediction Step:

$$\mathbf{x}_k = \mathbf{f}(\mathbf{x}_{k-1}, \mathbf{u}_k)$$

$$\mathbf{P}_k = \mathbf{F}_k \mathbf{P}_{k-1} \mathbf{F}_k^T + \mathbf{Q}_k$$

### 11.2 Update Step:

$$\mathbf{y}_k = \mathbf{z}_k - \mathbf{h}(\mathbf{x}_k)$$

$$\mathbf{S}_k = \mathbf{H}_k \mathbf{P}_k \mathbf{H}_k^T + \mathbf{R}_k$$

$$\mathbf{K}_k = \mathbf{P}_k \mathbf{H}_k^T \mathbf{S}_k^{-1}$$

$$\mathbf{x}_k = \mathbf{x}_k + \mathbf{K}_k \mathbf{y}_k$$

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_k$$

Here,  $\mathbf{x}$  represents the state vector,  $\mathbf{P}$  is the covariance matrix,  $\mathbf{K}$  is the Kalman gain, and  $\mathbf{Q}$  and  $\mathbf{R}$  represent process and measurement noise respectively [\[3\]](#)[\[9\]](#). The functions  $\mathbf{f}(\mathbf{x})$  and  $\mathbf{h}(\mathbf{x})$  are nonlinear functions, which are linearized using Jacobian matrices to apply the filter effectively [\[8\]](#)[\[10\]](#). These mathematical relationships allow accurate estimation of position and velocity in dynamic systems.

## 12. System Model and State Space Representation

The Extended Kalman Filter uses a system model to describe how the state of an object changes over time. The state vector includes position and velocity, such as  $(p_x, p_y, v_x, v_y)$  [2][5]. The model predicts the next state and updates it using sensor measurements from radar and lidar [3][9]. This representation helps in accurate tracking of moving objects in dynamic environments [6][10].

## 13. What-if Analysis in Sensor Fusion Systems

What if sensor data contains noise?

The system may produce inaccurate results, but filtering methods like the Extended Kalman Filter reduce noise and improve accuracy [2][5].

What if one sensor fails (radar or lidar)?

The system can still work using the available sensor, but accuracy may decrease compared to using both sensors [4][7].

What if measurements are incorrect or missing?

The prediction step helps estimate the state even without measurements, ensuring continuous tracking [3][9].

What if environmental conditions change (rain, fog)?

Radar performs better in harsh conditions, while lidar may be affected, so sensor fusion improves reliability [5][10].

What if system parameters are not tuned properly?

The estimation error may increase, so proper tuning of noise parameters is necessary for better performance [6][8].

## 14. Tools, Technologies, and Methods Used

Tools

- Visual Studio Code
- Debugging tools

- Compiler (MinGW)

#### Technologies

- C++ programming
- Sensor fusion
- Autonomous systems

#### Methods

- Kalman Filter
- Extended Kalman Filter
- State estimation
- Data fusion

## 15. Conclusion

This literature survey reviewed various techniques for object tracking in autonomous systems, focusing on sensor fusion and Extended Kalman Filter methods. The study highlighted the advantages of combining radar and lidar data for improved accuracy and reliability.

It also identified key limitations in existing approaches, such as handling nonlinearities and real-time implementation challenges. Based on these findings, the need for an efficient and robust tracking system is established, which forms the basis for the proposed work.

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